Global Unichip

#Initiation #AI #ASIC #DesignService

Buy
Initiation

Price: **NT$ 7,615** Target Price: **NT$ 12,000** Upside: **58%** Total return: **60%** Risk: High

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Bloomberg	3443 TT
RIC	3443.TWO
Market cap (USDbn)	32.1
Average daily T/O (USD|m)	337

Design Your Own Chips

The rapid proliferation of agentic AI workloads and persistently rising compute costs are prompting CSPs and AI labs to accelerate their in-house ASIC initiatives. We think GUC is a key beneficiary, as clients leverage its IP portfolio—particularly in advanced packaging—together with its turnkey services and unparalleled relationship with TSMC's DTP division. GUC's product portfolio is expanding from CPUs for Google and Microsoft to customized ASICs for many CSPs. We expect exceptional growth, supported by a fivefold rise in N3/5 wafer allocations in FY27E, indicating a high conversion rate from NRE to turnkey revenue. This growth outlook is underpinned by GUC's recent NT$65bn debt raise—equiv to 8x current cash—highlighting an asset-light company scaling working capital to support huge demand. Besides, GUC should benefit from operating leverage as sales surging 4x in FY27E (after doubling in FY26E) and may rise a further ~50% in FY28E. We forecast its EPS to surge 8x over FY26–28E, with our estimates nearly 200% ahead of Streets. We initiate GUC with a Buy at TP NT$12,000 on 30x avg-FY27/28E PER (35x FY27E). Risks are demand outlook and execution.

Rising tide for backend design services

Merchant AI GPUs (Nvidia, AMD) and custom ASICs (designed entirely by AVGO and MRVL) have dominated AI training/inference for the past several years, but hyperscalers and AI startups are increasingly moving to custom silicon (that is, Google's Customer-Own-Tooling (CoT) initiative) — designing main dies in-house, partnering with IP/APR (automatic place-and-route) vendors like GUC on back-end design (TK1) or production transfer (TK3), and outsourcing sub-dies/IOs to design houses. GUC is a key beneficiary of that CoT shift, thanks to a top management team drawn from TSMC's DTP division and expanding 2.5D/3D-stacking die-to-die (GLink) IP that is increasingly critical for large-reticle devices, where interposer APR design becomes essential.

Massive wafer allocation with tremendous optionality value

Our research indicates that GUC has secured 180k N3 and 50k N5 wafers for 2027, a step change from just 35k N3+N5 wafers in FY26E. We suspect that N3 projects cover Google's Axion CPU (50%+ of GUC's revenue in FY26E/27E), Maia 200/Cobalt 200, plus some hyperscale TK3 projects and TK1 workloads for a US AI networking startup. On N5, GUC will support a China auto, a China networking, and a US AI inference chip customer. GUC has stated it is engaging with all hyperscalers except Apple. We see the optionality in Google's TPU v10 (Icefish), custom silicon from AWS, Meta's Phoebe (MTIA 600), HBF from SanDisk/Hynix, and CPO — none of which is in our model. Should these convert, they could add NT$100+ EPS, most likely from 2028 onwards.

Net margin on the rise with rising TK1 mix and improving TK3 margin profile

Although GUC's gross margin may fall from 24% in FY25 to 19-20% in FY26E-28E on higher turnkey mix, operating leverage will deliver 200bps+ OPM expansion. GUC has also guided to a rising net margin over the next couple of years. We expect TK3 gross margin to grow from high single digits into the teens as more TK3 work involves SoIC/CoWoS, which carry far heavier APR workloads. TK1 mix should also rise from 20% in 2026 to 50%+ in 2027, a further margin support (higher design effort). GUC has 10+ TK1 tapeouts in the 2027 pipeline, from just a few in 2026.

KEY FINANCIAL AND VALUATION MATRIX

FYE Dec (TWD mn)	FY21A	FY22A	FY23A	FY24A	FY25A	FY26E	FY27E	FY28E
Revenues	15,108	24,040	26,241	25,044	34,141	73,926	337,530	481,701
Op. Profit	1,674	4,099	3,969	3,803	4,350	9,466	50,391	71,864
Net Profit	1,460	3,710	3,508	3,451	3,770	8,583	44,969	64,014
FD EPS (NT$)	10.9	27.7	26.2	25.6	28.0	63.9	332.1	472.2
Consensus EPS (NT$)	-	-	-	-	-	52.6	120.7	191.6
PER (X)	501.6	197.4	208.8	213.8	195.1	85.5	16.5	11.6
PBR (X)	139.1	90.9	75.6	65.3	56.9	38.9	12.8	8.2
Dividend Yield (%)	0.1	0.1	0.3	0.3	0.3	0.4	0.8	4.3
ROE (%)	30.0	55.7	39.5	32.9	31.1	54.1	117.3	86.4
ROA (%)	11.9	20.6	16.6	14.6	13.8	19.6	37.1	33.3

Source: Bloomberg, Aletheia/TAG

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Design Your Own Chips

We forecast GUC's revenue to compound at 100% over FY26E-28E, a sharp acceleration from 20% over FY21-25, on a broadening pipeline: Google's Axion 2 and 3, Microsoft's Maia 200/400 and Cobalt 200, Tesla's AI5/AI6.5, ADAS/robotics ASICs for China auto customers, networking and inference chips for AI startups, and further CSP ASIC/CPU work at AWS, Meta and Google. We also understand GUC is engaging on HBF with SanDisk and SK Hynix, and on CPO EIC layout and PIC COUPE and testing.

Merchant AI GPUs (Nvidia, AMD) and custom ASICs (designed end-to-end by AVGO and MRVL) have dominated AI training/inference for the past several years, but hyperscalers and AI startups are increasingly moving to custom silicon. Google's Customer-Owned Tooling (CoT) initiative is the clearest example: designing the main die in-house and outsourcing sub-dies/IOs to design houses such as MediaTek on the TPU v8t (Zebrafish) and v9t (Humufish), while leaving its inference variant, the TPU v8i (Sunfish), to be designed end-to-end by Broadcom across every tile. Our research suggests Google will carry this dual-track approach into its v10 variants. As architectures grow more complex, its focus should stay on the main die and potentially the sub-dies, while partnering with IP/APR (automatic place-and-route) vendors like GUC on back-end design (TK1) or production transfer (TK3) to offload the most time- and labor-intensive work.

We therefore see GUC as a key beneficiary of the CoT shift, thanks to a top management team drawn from TSMC's DTP division and expanding 3D-stacking die-to-die (GLink) IP that is increasingly critical for large-reticle devices, where interposer APR design becomes essential.

Furthermore, our research indicates that GUC has secured 180k N3 and 50k N5 wafers for FY27E — a step change from just 30k N3 wafers in FY26E. We suspect that the N3 projects cover Axion 2, Axion 3, Maia200 and Cobalt200, plus some hyperscale TK3 projects and TK1 workloads for a US AI networking startup. On N5, GUC will support a China auto customer and a US AI inference startup. It has stated it is engaging with all hyperscalers except Apple. We see the optionality in Google's TPU v10 (Icefish), custom silicon from AWS, Meta's Phoebe (MTIA 600), HBF from SanDisk/Hynix, and CPO — none of which is in our model. Should these convert, they could add NT$100+ to EPS, most likely from 2028 onwards.

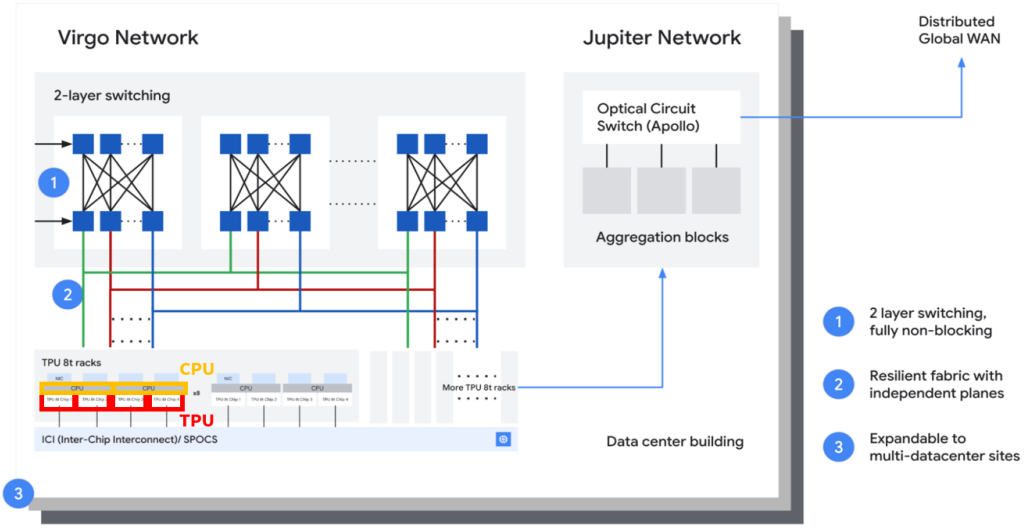
FIGURE 1: GUC'S MAJOR TK PROJECTS IN FY26E/27E

	TK	Node	Launch	Packaging	Chip volume (k units)	
					2026E	2027E
Google						
Axion 2	TK3	N3	3Q25	SoIC	1,160	5,000
Axion 3	TK3 + IP	N3	4Q27	SoIC	-	300
Tesla						
AI5	TK1	N3	2027	Flip Chip	-	800
Microsoft						
Cobalt 200	TK3	N3	2026	Flip Chip	240	200
Maia 200	TK3	N3	2026	CoWoS-S	-	1,200
Aspeed						
AST2600/2700	TK1	N28/N12	2025	Flip Chip	28,000	48,000
China Auto						
N5 Chip	TK1	N5	2026	Flip Chip	240	2,400
China Networking						
N5/6 Chip	TK1	N5/6	2027	Flip Chip	-	1,200
AI Startups						
N3 chip	TK2	N3	2026	CoWoS	-	500
N5 chip	TK1	N5	2026	Flip Chip	400	700

Source: The company, Aletheia/TAG

Google CPU Axion 2 and 3

FIGURE 2: GOOGLE TPU AI CLUSTER ARCHITECTURE



Source: The company, Aletheia/TAG

Google launched its Axion CPU (C4A) in late 2024, but we expect volumes will only take off from its second-generation Axion 2 this year. The driver for Axion 2 (N4A) this year is mainly to ship together with its TPUs. Its current TPUv7 (Ironwood, Ghostfish) ships together with Intel's CPU, with the TPU-to-CPU ratio at 2:1. However, we believe its TPUv8 families, including Sunfish from Broadcom and Zebrafish from MediaTek, will be equipped with Axion 2. We forecast combined Sunfish and Zebrafish shipments will reach 1.1mn this year. Total Axion shipments will reach 1mn+ this year accordingly and Axion 2 will reach 1-1.5mn units. Looking into 2027, we are forecasting total TPU shipments of 8.2mn, with 8mn from TPU v8 and v9 families. This implies demand of 4mn+ Axion CPUs for TPU racks. In fact, we expect a total of close to 5mn Axion CPU, including 4-4.5mn Axion 2 and 500k+ Axion 3 (Axion Next).

GUC management confirmed that Axion 2 is a single-die device (likely referring to compute die). It began ramping around 3Q25, giving only four months of contribution in 2025 at a percentage share of revenue in the teens. In 2026 the project is contributing a full 12 months at a higher monthly output, taking its revenue share to above 50%, and demand visibility extends into 2027. Management has previously flagged that TSMC wafer allocation is the key bottleneck and does not expect to meet 100% of customer demand. However, our research suggests GUC has lifted its FY27 N3 allocation from TSMC to 180k wafers, from 110k originally, covering not only Google Axion CPU demand but other N3 projects such as AI5, Cobalt200 and Maia200. Axion 3, again on N3 like Axion 2, should come in at a higher ASP — potentially double — with turnkey revenue starting at the early FY27E and becoming meaningful in 2H27.

Management noted that Axion 2 and 3 projects are both TK3 engagements, which leaves limited room to negotiate higher margins despite the strong volumes; instead, it intends to leverage that track record to win accelerator business.

FIGURE 3: SUPPLIES OF CPU

	2025	2026E	2027E	Remarks
Nvidia	2.2m	~5.2m	6m	NVDA targets 6m units of Vera CPU production in '27, up from ~2.5m in 2H26
AMD	6.5m	Up to 9m	Up to 15m	Demand could be as high as 18m units in '27. Recently secured N2/3 wafers from TSMC for 2027 productions. Working on CoWoS supply.
Intel	14m	~16m	~21m	Diamond Rapids delayed (twice) to '27. Clearwater Forest to be shipped from mid-'26.
Graviton	1.8m	~2.5m	~3.5m	Graviton 5 is a competitive device (192 cores on TSMC's N3)
Axion	<0.1m	~1m	~4.5m	Google to use Axion2 in the TPUv8 rack, replacing Intel's CPU. For agentic AI, Google to still focus on x86 CPUs (~85%).
Total	24.7m	33.7m	50m	CPU volume to double in three years (among top-five suppliers), with AMD and Google growing at the fastest pace.

Source: Aletheia/TAG

FIGURE 4: SPECIFICATION OF SELECTED CPUS FOR AI COMPUTES

Specifications	AWS – Graviton5	Google – Axion2	ARM – AGI CPU	Nvidia – Vera
Architecture	ARM – N3	ARM – N3	ARM – N3	ARM – Olympus
Process node	N3	N3	N3	N3
Cores	192	72	136	88
Peak speed	3.7GHz	3.5GHz	3.7GHz	3.5GHz
TDP	300W	300W	300W	250W
SRAM cache	~400MB	200MB	640MB	338MB
Mar DRAM	4TB	2TB	6TB	1.5TB
Max bandwidth	~1,100GB/s	~1,000GB/s	1,500GB/s	1,200GB/s
Release	1H27	2H26	2H26	2H26

Source: The company, Aletheia/TAG

FIGURE 5: GOOGLE HARDWARE

System level	Cypress V8t	Ironwood	Cypress V8i	GB300 NVL72	GB300 SuperPod	VR SuperPod
TPU/GPU name	Zebrafish V8t	Ghostfish	Sunfish V8i	Blackwell	Blackwell	Rubin
Total TPU or GPU /Pod	9,600	9,216	1,152	72	576	1,152
Racks	150	144	36	1	8	16
Tray	16	16	8	18	18	18
TPU or GPU/rack	64	64	32	72	72	72
Device level	1xN3P+1xN4P	2xN3E+1xN5	2xN3P+1xN4P	2xN4B	2xN4B	2xN3P+1xN3P+1xN4P
HBM/TPU or GPU	216GB	288GB	288GB	288GB	288GB	288GB HBM4
SRAM/TPU or GPU	128MB	192MB	384MB	192MB	192MB	256-320MB
TPU/GPU per tray	4x V8t	4x V7	4x V8i	2x2	2x2	2x2
CPU	2x Axion	1x Xeon	2x Axion	2x Grace	2x Grace	2x Vera
FP4 (Eflop/pod)	121	-*	11.6	1.8	14.4	57.6
FP8 (Eflop/pod)	60.5	42.5	5.8	0.9	7.2	28.8
FP4/TPU (Pflop)	12.6	-*	10.1	25	25	50
FP8/TPU (Pflop)	6.3	4.6	5.0	12.5	12.5	25
System TDP (MW)	~10.5	12.5	~1.1	0.14	1.25	3.2
TDP Per rack (KW)	~70	87	~30.6	140	140	~230
TDP per device* (KW)	~1,094	1,356	~955	1,944	2,170	2,778
Workloads	AI Training	General purposes	Inference/agentic	General purposes	General purposes	General purposes

Note: *Ironwood/TPU V7 doesn't support FP4. *TDP per device (TPU & GPU) takes into the considerations of all the associated devices go into the system i.e. CPU, NIC, switches, HBM, DRAM, PMIC etc. Source: NVDA, GOOG, Aletheia

Google TPU v10

Google's TPU v10 (Icefish) platform will likely require 3D/3.5D packaging-related IP, given its use of TSMC's SoIC-X platform — an area GUC management believes plays to the company's strengths. GUC's addressable scope also extends to physical back-end design and to SerDes and UCIe/HBM stacking IP, should the customer require it.

For instance, GUC's IGAD2DZ05A and IGAD2DZ05A are both GLink-3D high-speed Die-to-Die interface PHY IPs. Each IP is used to transmit data between dies through TSMC's SoIC-X 3D stacking technology (3DFabric), which supports CoW assembly, with face-to-face die stacking. That said, our research indicates that GUC is engaging with Google on this project, but while we await greater clarity, our forecasts through 2029 do not include any TPU v10 contribution.

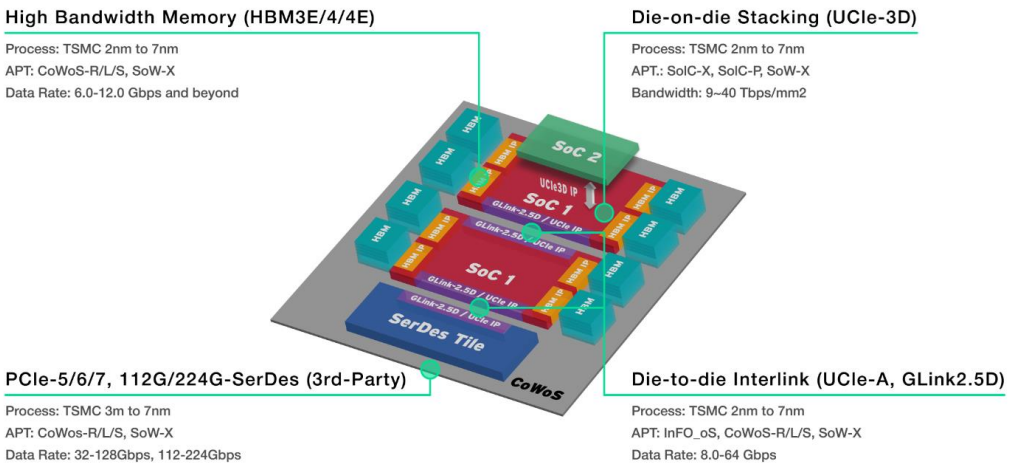
That said, we believe MediaTek remains the frontrunner to win the TPUv10, if it has not already done so. We expect Google to take care of the compute tiles and uncores (sub-compute) by itself and work with IP/APR vendors like GUC for backend design, and outsource the 6x IOs, HBM base-die and 3.5D packaging to MTK. This compares with 4x IOs, 4x uncores and 2.5D packaging for the TPUv9.

FIGURE 6: GUC'S DIE-TO-DIE 3D GLINK IP PLATFORM

	TPU V8		TPU V8.5	TPU V9		TPU V10	
Variants	Inference (Zebrafish)	Training (Sunfish)	Sunfish+	Inference	Training	Inference	Training
Compute	AVGO	GOOG	AVGO	GOOG	GOOG	GOOG	GOOG
Logic/Base	-	-	-	MTK	MTK	GOOG	AMD/GOOG
I/O	MTK	AVGO	AVGO	MTK	MTK	MTK	AMD/AVGO
3DIC	-	-	-	-	-	GUC	AMD/AVGO
HBM Procurement	GOOG	AVGO	AVGO	GOOG	GOOG	GOOG	GOOG
HVM	4Q26	2H26	2H27	1Q28	1Q28	1Q29	1Q29

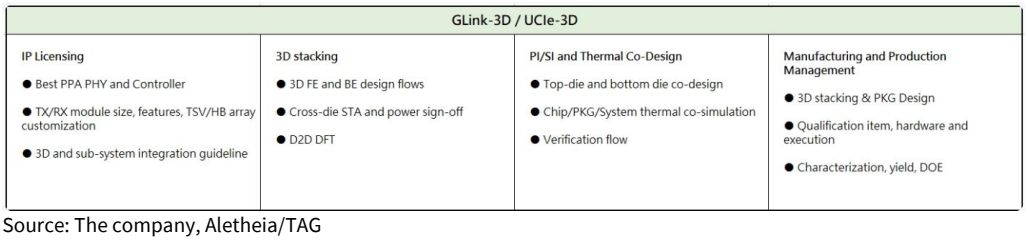
Source: Aletheia/TAG

FIGURE 7: GUC ADVANCED PACKAGING TECHNOLOGY (APT) PLATFORM



Source: The company, Aletheia/TAG

FIGURE 8: GUC'S DIE-TO-DIE 3D GLINK IP PLATFORM



Source: The company, Aletheia/TAG

Tesla AI5/AI6.5

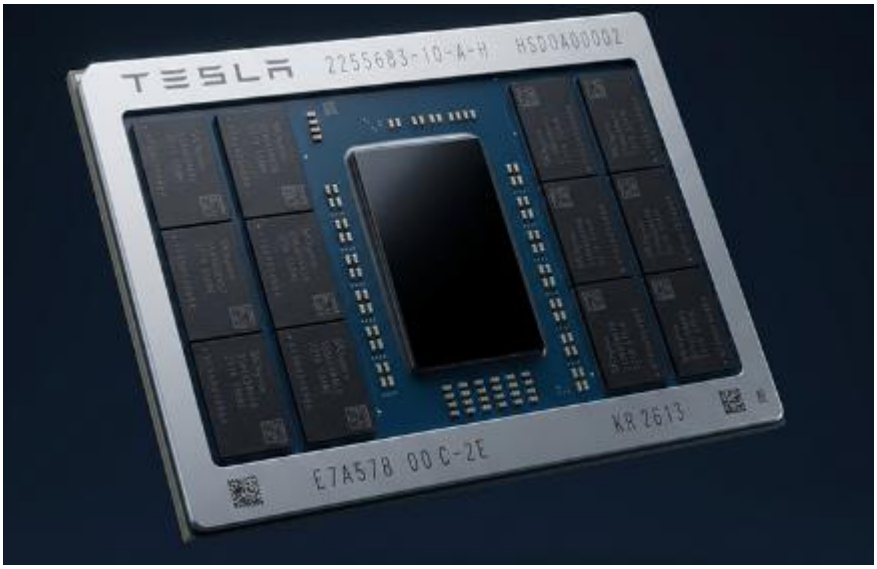
Tesla said earlier this year that AI5 production is planned for 2027 and AI6 for 2028. Tesla's AI silicon is a scalable design that can be scaled down for use in humanoid robots and self-driving cars but also scaled up for AI data center usage. For the time being, it plans to initially deploy AI5 for the Optimus humanoid robot and self-driving.

GUC management indicated that the chip taped out in 2Q26, with turnkey revenue starting in 2H27 and continuing through 2028, consistent with the typical one-year tape-out-to-production lead time for a TK1-plus-back-end-design engagement. On the roadmap, Samsung will handle the AI6, while the AI6.5 returns to Taiwan. Note that the Tesla AI5 is dual sourced from TSMC's N3 and Samsung's N2, with TSMC carrying the bulk of AI5 volume. Our checks point to AI5 substrate orders of 700-800k units in FY27E, with front-end wafer volumes

higher still at over 1mn. Packaging-side checks indicate materially lower volumes, which we attribute to Tesla allocating work across multiple OSATs.

GUC serves Tesla as a TK1 vendor, handling back-end design, packaging/testing and wafer placement. Management indicates a TK1 gross margin of 15%-20%. We expect the Tesla AI5/6.5 to stay at a single digit percentage of GUC revenue next year before ramping above 40% in 2028, as AI5/6.5 volumes scale past 5mn units.

FIGURE 9: TESLA AI5 DEVICE



Source: The company, Aletheia/TAG

China Auto

A key China automotive chipmaker is expected to account for a single-digit share of revenue this year, rising to double digits next year. Its chips address both automotive and robotics applications. GUC serves this customer as a TK1 vendor.

The current generation of the device is fabricated on TSMC's N5 node, with turnkey revenue starting in 2H26 and continuing through 2027 and 2028; management expects a higher fulfilment rate on this project than on its wafer-constrained programmes.

GUC has also secured the next-generation product on N3, which is scheduled to tape out by end-FY26E. Margins here are higher than on the Google CPU projects and above the typical 15-20% TK1 range; we estimate the gross margin will reach 30%. We expect this customer to move from 7% of revenue in FY26E to 12% in FY27E, and from 20% of bottom-line contribution to ~30% — making it the second-largest profit contributor to Google Axion CPU.

Microsoft Maia200/400

Microsoft presented the Maia200 architecture at Hot Chips this August. The part is a single 820mm² die on TSMC 3nm, with 140bn transistors, six HBM3e stacks, a 750W TDP, 7 TB/s of memory bandwidth and 10,000 TFLOPS of FP4.

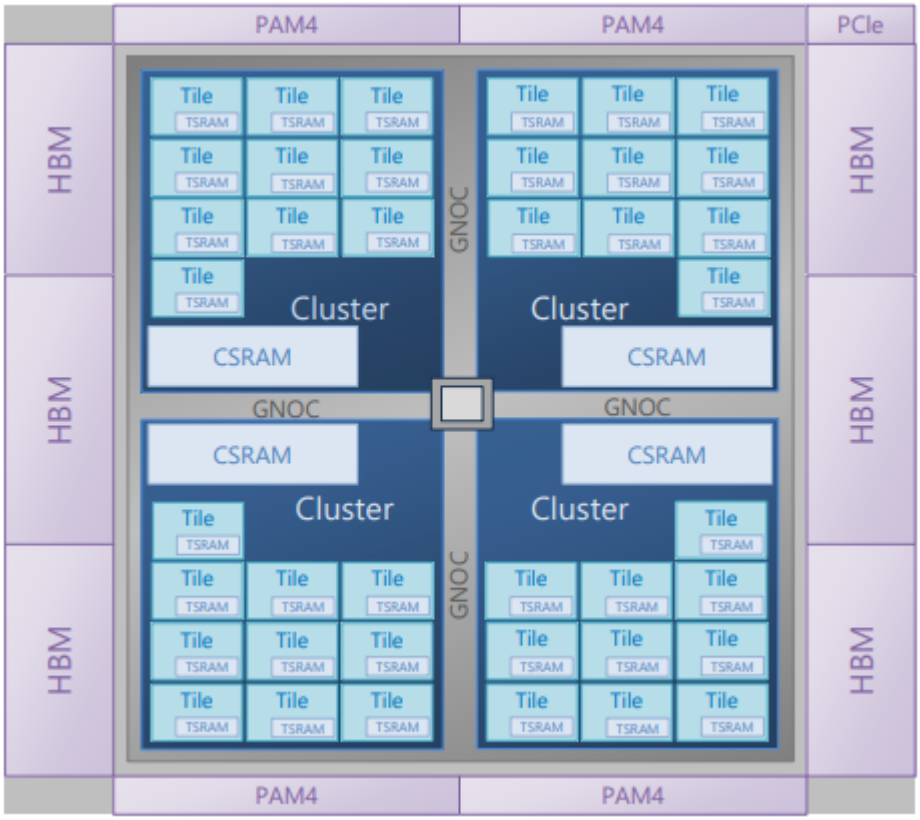
On Maia 100, Microsoft worked without a backend design partner: it licensed SerDes directly from a third party rather than through Broadcom or Marvell, submitted the design straight to TSMC on N5 and designed the package itself, with GUC involved only in supply chain management (TK3) rather than in design (TK1).

That has loosened with Maia200, a Microsoft-architected SoC that brings in GUC (TK3), Marvell and TSMC on N3 as partners, giving Microsoft economies of scale it would struggle to reach in-house. Each Maia 200 blade server holds four accelerators alongside a single Cobalt 200 Arm CPU. That is a 1:4 CPU-to-XPU ratio — the same mix AWS chose for Trainium 3 Gen2. Cobalt 200 is TSMC 3nm with 132 Neoverse V3 cores, announced at Ignite 2025.

We estimate Cobalt 200 volume will be 240k/600k in 2026E/27E and Maia200 to be 1mn+ in 2027E. Both run through GUC's TK3 engagement model.

Looking ahead, although Marvell won the Maia300 design, we see a high probability of GUC regaining Maia400, and we understand GUC is already engaged in the Maia400 design.

FIGURE 10: MICROSOFT MAIA 200 SOC



Source: The company, Aletheia/TAG

AI startups

GUC is also working with several AI startups through its TK1 and TK2 models. These customers typically focus only on front-end design and have limited experience in physical design implementation or in supply chain management, including packaging and test deployment.

At the time of writing, we expect a US AI inference startup to take 400k and 700k units of ASICs on TSMC's N5 process node under GUC's TK1 engagement model.

Looking into 2027E, GUC should add another US AI networking startup on TSMC's N3 process node under its TK2 engagement model, with initial volume of 500k in 2H27.

Meta Phoebe (MTIA600)

Meta has been slow in bringing out its in-house ASICs at scale over the past two years as it still relies heavily on Nvidia GPUs and Google TPUs. We currently expect Meta's AI ASICs, including MTIA 400 (Iris+), MTIA 450 (Arke), and MTIA 500 (Astrid), to remain at an annual chip volume scale of 300-600k.

Into 2028, we expect Meta to begin development of MTIA600 (codenamed Phoebe), built on TSMC's N2P/N3/N3E nodes using SoIC and CoWoS-L. Our research suggests GUC is engaged with Meta under a TK3 business model.

Our model does not yet factor in any contribution from Meta ASICs. If the project proceeds, we estimate annual earnings power of ~NT$25, assuming 300k units at a TK3 margin profile.

We also see a reasonable chance GUC engages under a TK1 model, given its stated intent to cover both front-end and back-end design and mass production of the I/O dies, as well as integration of the Rivos-designed compute die. Under that scenario, the earnings contribution would be materially higher.

AWS CPU Graviton 6

The Graviton CPU is a family of 62-bit ARM-based CPUs designed by Annapurna. AWS announced that Trainium 3 will use Graviton as head nodes at 1 CPU to 4 XPUs, starting with Graviton 4 and moving to Graviton 5 for future clusters. In this design the Graviton 4 sits on the liquid-cooled compute tray alongside the Trainium 3 modules and the PCIe 6.0 switch, mirroring how Nvidia places Grace and Vera on the same tray as the GPUs. AWS is expected to offer only the Graviton version of the 144-chip SKU, reserving x86 for the smaller configuration.

We are aware that GUC is engaging with AWS on CPU projects. This is very likely pointing to Graviton 6. The recent cadence is a clean two-year drumbeat of its CPU iterations: Graviton 3 in Dec 2021, Graviton 4 in Nov 2023, Graviton 5 in Dec 2025. That points to a Graviton 6 spec to settle in 2H27 and general availability around mid-2028.

We have yet factored in the contribution from AWS Graviton CPU in our model. If this comes through, we believe it is likely to contribute annual earnings power of ~NT$40, assuming 1mn units at a TK3 margin profile.

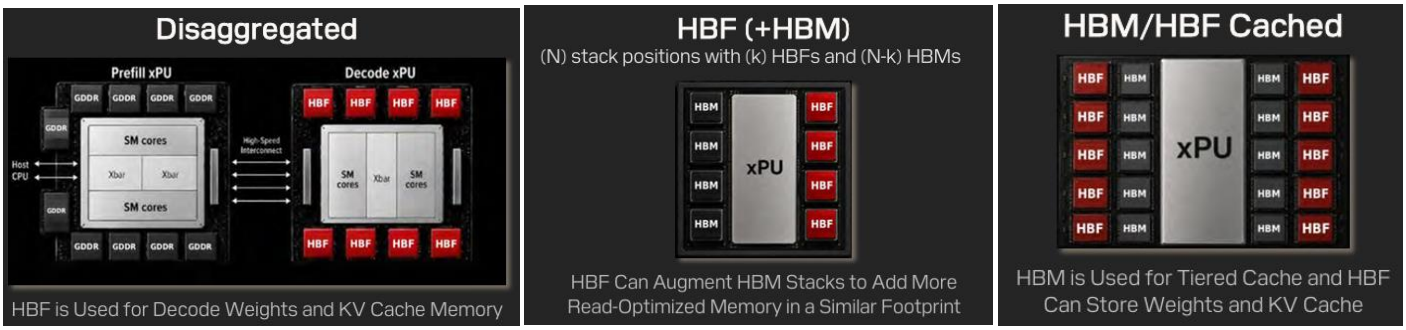
HBF Memory

GUC's management has publicly stated that it is engaging with key memory makers and foundry partners on HBF projects. Historically, GUC has had limited exposure to memory-related business — most HBM base-die work, for instance, sits with memory vendors' in-house solutions. It does, however, hold 3D stacking die-to-die IP for memory stacking. On HBF, we believe GUC is in early-stage engagement with SanDisk and SK Hynix on the stacking APR. Even so, we suspect HBF is still too early to offer near-term revenue visibility. While the SSD controller business shows signs of having resumed (with SK Hynix, we believe), the revenue contribution is likely small — low- to mid-single digits in 2026 — relative to GUC's current revenue base.

At SanDisk (SNDK) Investor Day this August, management noted that the key milestone was the tape-out of SNDK's first HBF die, with customer samples targeted for 2027 and potential production in 2028. This suggests HBF has progressed beyond concept development and standards formation into early silicon execution, making it one of the more tangible emerging memory architectures within SNDK's AI data center portfolio. In addition, SNDK announced that Tenstorrent and Meta have joined the HBF OCP initiative, alongside existing partners like SK Hynix and Google.

The concept behind HBF is similar to that of HBM: both are designed to stack memory—NAND flash in the case of HBF and DRAM in the case of HBM—close to compute devices such as GPUs and ASICs. However, HBF offers materially higher density at a much lower cost structure, addressing HBM's scalability limitations, even as HBM remains superior in speed and bandwidth. If HBF proves commercially viable, we believe it is likely to complement, rather than replace, HBM through mixed HBM/HBF configurations in the same package alongside logic devices, while working with eSSDs to create a denser and more cost-efficient near-compute memory layer for inference workloads.

FIGURE 11: HBF ARCHITECTURE PROPOSALS



Source: SNDK Investor Day (August 2026)

The use case is particularly relevant for long-context and multi-agent AI inference, where token generation increasingly depends on large, fast-access memory pools for model serving and KV cache. SNDK showed that a fully developed HBF-supported device with 4TB of density could deliver comparable token output using only half the number of GPUs required in an HBM-only configuration with 192GB of memory. This points to potential savings in GPU count and power consumption, lower total cost of ownership, and meaningful capex efficiency gains. If SNDK's claims prove accurate, the center of gravity in AI compute could shift further toward memory and away from GPUs or ASICs, at least for inference. This may also help explain why NVIDIA is not part of the HBF OCP initiative despite the apparent advantages.

FIGURE 12: HBF VS HBM PERFORMANCE COMPARISON FOR AI TOKEN GENERATIONS



Source: SNDK 2026 Investor Day (August 2026)

CPO EIC and PIC COUPE

GUC has publicly confirmed its participation in CPO development but has not disclosed the scope of engagement or the business model. Our checks indicate GUC is working with Lightmatter and Ayar Labs, several large customers where terms are not yet final and ATE vendors Advantest and Teradyne to assemble end-to-end back-end support spanning COUPE back-end APR and testing/packaging supply chain management.

Given that CPO packaging and testing remain nascent and complex, we see a high likelihood of GUC winning a turnkey mandate, spanning both PIC for the COUPE process and EIC for tapeout and APR. That said, we believe any meaningful CPO revenue contribution is a mid-term story.

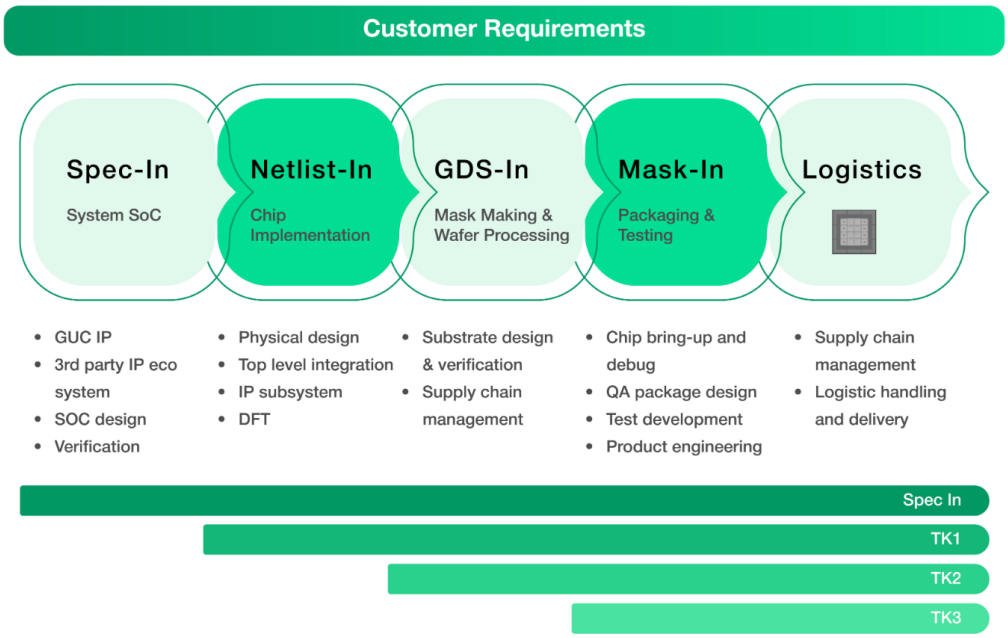
GUC's business model

According to GUC's own framing, Broadcom, Marvell and MediaTek are primarily function- and spec-design houses — SoC front-end design, effectively a "TK0" scope — and in which they are clearly more capable than GUC. Hyperscale CSPs nonetheless want to own that development themselves, outsourcing to these design houses only where they cannot get past a specific technical hurdle. That outsourcing is expensive: ASIC design houses run gross margins of roughly 40%-80%.

The consequence is that CSPs will progressively bring front-end design in-house up to the limit of their own capability and then tape out at TSMC. The back-end flow, however, is neither something TSMC provides nor something CSPs can staff economically — automatic place-and-route (APR) alone can absorb several hundred engineers per project, on top of a long tail of ancillary engineering work.

That is precisely GUC's positioning: weaker than the design houses in front-end design, but with a widening integration capability, which fits a CSP that keeps its core technology in-house and outsources the back end. The division of labor is unambiguous. Looking further ahead, ASIC design service providers such as GUC need not be confined to today's scope; they can take a program on a full-turnkey basis, with the CSP allocating individual engineering stages — back-end implementation and integration, for example — lowering the CSP's costs without compromising its in-house development principle.

FIGURE 13: GUC'S ENGAGEMENT MODELS



Source: The company, Aletheia/TAG

GUC categorizes its design effort into TK count as illustrated in the figure above. TK1, TK2, and TK3 do not refer to different chip generations, nor are they three sequential stages that every chip must go through. Rather, they indicate the point of project maturity at which a customer hands the program over to an ASIC design service provider.

GUC itself is best characterized as an ASIC design services and turnkey production company rather than a pure-play IP vendor such as Arm or Synopsys. It may develop, license, and integrate IP, but its core value proposition is turning a customer's ASIC design into a product that can be manufactured, packaged, tested, and produced at scale.

The lower the TK number, the earlier the handoff, the deeper the engineering involvement, and typically the higher the value-add—and margin profiles—for the ASIC design services provider:

- TK1: The customer provides the netlist, while the service provider takes responsibility for the entire physical implementation flow through mass production.
- TK2: The customer provides the GDSII database, with the service provider managing tape-out, packaging, testing, qualification, and volume production.
- TK3: The product design and manufacturing flow are largely established, and the service provider takes over volume production and supply-chain management.

TK1 (Netlist-in): By the netlist stage, the customer has typically already completed chip architecture and algorithm development, RTL design, functional verification, IP selection and integration, and logic synthesis to generate the gate-level netlist. However, the netlist only specifies (1) the standard cells, memory blocks, macros, and IP blocks used in the design; and (2) how these components are logically interconnected. It does not yet determine their physical coordinates on the die, the power-delivery network, or how the metal interconnects are routed – GUC handles all of this, converting the netlist (normally the pin assignment is fixed) into a file called GDS (GDS-In), which TSMC's foundry process team then uses to fabricate the product wafers.

After taking over the project, ASIC design service companies like GUC perform:

(1) die-size estimation and floor planning;

(2) placement of macros such as CPUs, SRAM, SerDes, and HBM PHYs;

(3) power-grid and bump-map planning;

(4) standard-cell placement;

(5) clock-tree synthesis;

(6) signal routing;

(7) scan and DFT integration;

(8) Static timing analysis;

(9) IR-drop, electromigration, and signal-integrity analysis;

(10) physical verification, including DRC, LVS, and antenna checks;

(11) engineering change orders and timing closure; and

(12) generation of the final GDSII for delivery to the foundry.

Therefore, although TK1 does not involve designing the chip architecture from scratch, it encompasses the most challenging stage of physical implementation, that is, turning a logical circuit design into a manufacturable chip layout.

The physical implementation of large AI ASICs on advanced process nodes is particularly challenging. These chips may be multiple times the reticle-size limit, contain tens of billions of transistors, and consume several hundred watts to more than one kilowatt. Achieving timing, power, and routing closure alone can require a large engineering team and substantial EDA computing resources.

This is the core competency of physical back-end design, with a particular focus on optimizing die size, timing convergence, and manufacturing yield.

TK2 (GDS-in): In a TK2 engagement, the customer has already completed the physical implementation, either internally or through another design house, and provides GUC (or a peer such as Alchip) with a sign-off GDSII database.

This implies that:

- On-die transistor placement and metal routing are largely finalized.
- The ASIC service provider is no longer responsible for the primary place-and-route (P&R) implementation.
- Its ability to optimize on-die power, performance, and area (PPA) is therefore limited.
- If a fundamental design issue is identified, the customer generally needs to revise the GDSII and, in many cases, perform a re-tape-out.

TK2, however, should not be interpreted as merely a manufacturing handoff. Depending on the project scope, the ASIC service provider may still be responsible for a broad range of post-design engineering and manufacturing execution activities, including:

- Foundry tape-out coordination and mask preparation;
- Wafer starts and foundry capacity allocation;
- Package architecture and substrate design;
- Interposer, redistribution layer (RDL), and bump routing;
- Signal integrity (SI), power integrity (PI), and thermal simulations;
- Probe card, load board, and test socket development;
- Wafer sort (CP) and final test (FT) program development;
- Silicon characterization and debug;
- Product qualification and reliability validation; and
- Yield improvement, production ramp, and supply-chain management.

For conventional monolithic 2D devices, TK2 typically contains significantly less engineering content than TK1. However, for advanced packages, including CoWoS, HBM-based designs, and chiplet architectures, the packaging and manufacturing engineering workload under TK2 can remain highly substantial. In many leading-edge AI/HPC programs, package implementation, SI/PI optimization, thermal co-design, and test development have become critical engineering disciplines, making TK2 considerably more value-added than its traditional definition would suggest.

TK3 (Production transfer): TK3 typically occurs after the product design, packaging, testing, and even the production flow has already been established, with manufacturing execution subsequently transferred to the ASIC service provider.

The transfer package may include:

- GDSII and mask-related data;
- Wafer-process and manufacturing specifications;
- Package BOM, substrate, and assembly data;
- Wafer-sort and final-test programs;
- Probe cards, load boards, and sockets;
- Qualification and reliability data;
- Golden samples;

- Yield history and binning rules; and
- Foundry and OSAT supply-chain and quality-control processes

TK3's core value lies not in chip design, but in transferring an already manufacturable product into a different production and supply-chain system while maintaining stable and efficient volume manufacturing.

Accordingly, TK3 requires the least upfront NRE design work, but this does not mean it is operationally straightforward. Transferring an advanced-node product still requires test correlation, yield validation, supplier qualification, change control, quality tracking, and capacity management.

GUC's website describes TK3 as "mass-production introduction and transfer", rather than simply "GDS-in." The exact maturity and scope of the transferred deliverables may vary depending on the customer contract.

Largest funding in the company's history for tremendous growth potential

GUC's board approved on 19 August – wo separate resolutions filed to MOPS the same evening, together worth roughly NT$65.5bn (~$2.1bn), the largest capital raise in the company's history: (1) syndicate loan – NT$40bn size, with the company able to adjust the total facility within a±25% range — so up to NT$50bn if upsized. The tenor is five years from the first drawdown date and the proceed is used to bolster medium-to long-term working capital; and (2) ECB – this is GUC's first overseas unsecured convertible bond issuance worth of NT$25.5bn (~$800mn). The proceed will be used for foreign-currency materials procurement.

The underlying driver is the ASIC design service business model's cash cycle. ASIC design service providers advance payments on behalf of customers for wafer starts, packaging and testing, and materials, while collecting later — creating a working capital timing gap that widens as order scale and project count grow. Because wafer, CoWoS and material costs are USD-denominated, funding them with a USD instrument is a natural hedge; the NTD loan covers general operating needs.

Cloud computing, AI training and cloud connectivity together made up nearly 80% of 2Q26 revenue, with CPU and AI inference projects for two US CSPs due to enter mass production between late this year and early next.

GUC's 2Q26 contract liabilities hit NT$13.21bn (+63% QoQ), indicating strong customer prepayments matched to the wafer start and material prep schedule. Management said on the July earnings call that next-year capacity allocation would be confirmed within one to two months and that they were "very, very optimistic" about securing it — the raise is widely read as prepayment for goods and locking in advanced packaging capacity.

Notably, GUC has long maintained an exceptionally strong balance sheet, with a net cash ratio of 70-90% and cash and short-term investments of NT$8bn as of end-2025. That base alone has supported consistent earnings of NT$25-30 EPS per year over the past five years — and it did so while many of GUC's NRE projects carried a lower NRE-to-turnkey conversion rate than we expect from this year onward, as conversion should step up sharply over the next few years given that most of its engagements are with large system houses such as hyperscale CSPs. We think the NT$65-66bn of combined syndicated loan and ECB proceeds — 8x its current cash level — signals a business scaling to 5-10x its present revenue and profit; this is a further support for our FY27E-28E estimates, which sit ~200% above consensus and imply ~4x topline growth and 6-7x bottom-line growth over FY26E-28E.

Earnings forecast and assumptions

Management guidance for FY26E: Management guides for FY26E revenue to nearly double YoY, on double-digit NRE growth and several-fold growth in turnkey as TSMC wafer allocations increase. For 3Q26, revenue is guided to grow by the low-teens QoQ, led by turnkey, with NRE down by the low-teens QoQ. The mix shift implies a slight QoQ decline in GPM, while opex should be flat QoQ. Revenue is then guided to a record high in 4Q26, growing faster than in 3Q26.

Our assumptions for FY26E: We expect revenue growth of 23% QoQ in 3Q26, accelerating to 85% QoQ in 4Q26 as Axion 2 and China auto begin to ramp meaningfully from 4Q26. That translates into 112% YoY revenue growth for the year. We expect the gross margin to decline 50bps QoQ in 3Q and 40bps QoQ in 4Q as TK3 projects ramp. We do, however, expect the net margin to stay resilient at its 2Q level of 11% or better on flat opex (NRE is labour-intensive while turnkey is far less so, which drives strong operating leverage).

Management guidance for FY27E: For FY27E, GUC expects revenue growth to accelerate further from FY26E, implying more than double YoY. Opex growth should be lower than in FY26. On the project pipeline, management did not rule out engagements with Microsoft on Maia 400 and with Meta on MTIA ASICs.

Our assumptions for FY27E: We expect FY27E revenue to grow by 357% vs the consensus of ~100%, driven by Axion 2/3 CPU, which remains the largest contributor at 50%+ of revenue mix as in 2026. We expect total Axion CPU series units to rise from 1mn+ in FY26E to 5mn+ in FY27E, with GUC's ASP doubling YoY. China auto and the US AI startups should also ramp through 2027. For the Tesla AI5 we assume ~800k units based on our substrate and front-end checks, which point to ~1mn units of allocation at TSMC. We do not rule out the chance of further upward revisions once the outstanding engineering issues are resolved and Tesla sees more demands for its AI5. Net-net, we think GUC's 180k N3 and 50k N5 wafer allocation is sufficient to support our above-consensus estimates.

Declining gross margin (more turnkey mix) but expanding operating margin on operating leverage as well as improving TK3 margin and rising TK1 mix: Management guides for a TK1 gross margin of around 15-20%, and as high as 30%+ in some cases, TK2 to a mid-teen percentage and TK3 to roughly 10%. We see this mix as a major source of the discrepancy in sellside estimates for GUC's FY27E bottom line. We expect several key projects to come in at the high end of the TK1 margin range, including China auto, China networking, and US AI inference/networking startups.

On a blended basis, we also expect TK3 margins to move gradually from high-single digits to the low teens, so the dilution from a larger TK3 mix should be smaller than the market expects (some assume a mid-single-digit TK3 gross margin, which we think reflects outdated assumptions drawn from flip-chip and legacy packaging devices). With more TK3 projects built on complex chiplet architectures and using SoIC/CoWoS packaging at large reticle sizes, we believe the APR work becomes harder and should command higher margins.

FIGURE 14: GUC P&L FORECAST VS CONSENSUS

	3Q26E			4Q26E			2026E			2027E			2028E		
(TWD$mn)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)	Aletheia	Street	Var (%)
Sales	17,030	15,661	8.7	31,551	19,335	63.2	73,926	60,180	22.8	337,530	134,115	151.7	481,701	207,992	131.6
Gross profit	3,579	3,240	10.5	6,504	3,949	64.7	16,190	13,284	21.9	69,261	26,340	162.9	93,654	39,231	138.7
OP profit	2,058	1,912	7.6	3,908	2,547	53.4	9,466	7,853	20.5	50,391	19,101	163.8	71,864	29,459	143.9
Net profit	1,865	1,669	11.8	3,516	2,233	57.5	8,583	7,025	22.2	44,969	16,649	170.1	64,014	25,846	147.7
Diluted EPS (TWD$)	13.90	12.26	13.3	26.20	16.69	57.0	63.94	52.64	21.5	332.14	120.74	175.1	472.21	191.57	146.5
Gross margins (%)	21.0	20.7	0.3	20.6	20.4	0.2	21.9	22.1	(0.2)	20.5	19.6	0.9	19.4	18.9	0.6
OP margins (%)	12.1	12.2	(0.1)	12.4	13.2	(0.8)	12.8	13.0	(0.2)	14.9	14.2	0.7	14.9	14.2	0.8
Net margins (%)	11.0	10.7	0.3	11.1	11.5	(0.4)	11.6	11.7	(0.1)	13.3	12.4	0.9	13.3	12.4	0.9

Source: Bloomberg, Aletheia/TAG

FIGURE 15: GUC'S PROJECT-BASED MODEL

Project	TK	Core Node	KGD (k)	Revenue (US$ k)	Revenue Mix (%)	GPM	EPS
2026 key projects (A)				**1,837**			**36.66**
Axion2	TK3	N3	1,100	1,100	63%	10%	18.18
Cobalt200	Tk3	N3	240	192	11%	12%	3.87
AST2600	TK1	N28	24,000	168	9%	18%	5.64
AST2700	TK1	N12	4,000	52	3%	18%	1.75
US AI inference startup	TK3	N5	406	365	20%	12%	7.36
China Auto	TK1	N5	240	120	7%	30%	7.25
Core business + Others (B)							**27.28**
Total 2026 EPS (A) + (B)							**63.94**
2027 Key projects (A)				**10,079**			**276.40**
Axion2	TK3	N3	5,000	5,000	50%	10%	83.94
Axion3	TK3+IP	N3	300	600	6%	12%	13.43
Maia200	TK3	N3	1,200	720	7%	9%	11.28
Cobat200	TK3	N3	200	160	2%	9%	2.51
AI5	TK1	N3P	800	1,200	12%	22%	52.38
AST2600	TK1	N28	39,000	273	15%	18%	9.47
AST2700	TK1	N12	9,000	117	6%	18%	3.93
China Auto	TK1	N5	2,400	1,200	12%	30%	73.87
US AI networking startup	TK2	N3	500	250	2%	13%	6.16
US AI Inference startup	TK1	N5	700	560	6%	18%	19.43
China Networking	TK1	N5	1,200	300	3%	30%	18.47
Core business + Others* (B)							**55.74**
Total 2027 EPS (A) + (B)							**322.82**

Source: Aletheia/TAG (*) We expect the spillover effect of some of the 2026 key projects into 2027 revenue recognition

Valuation and recommendation

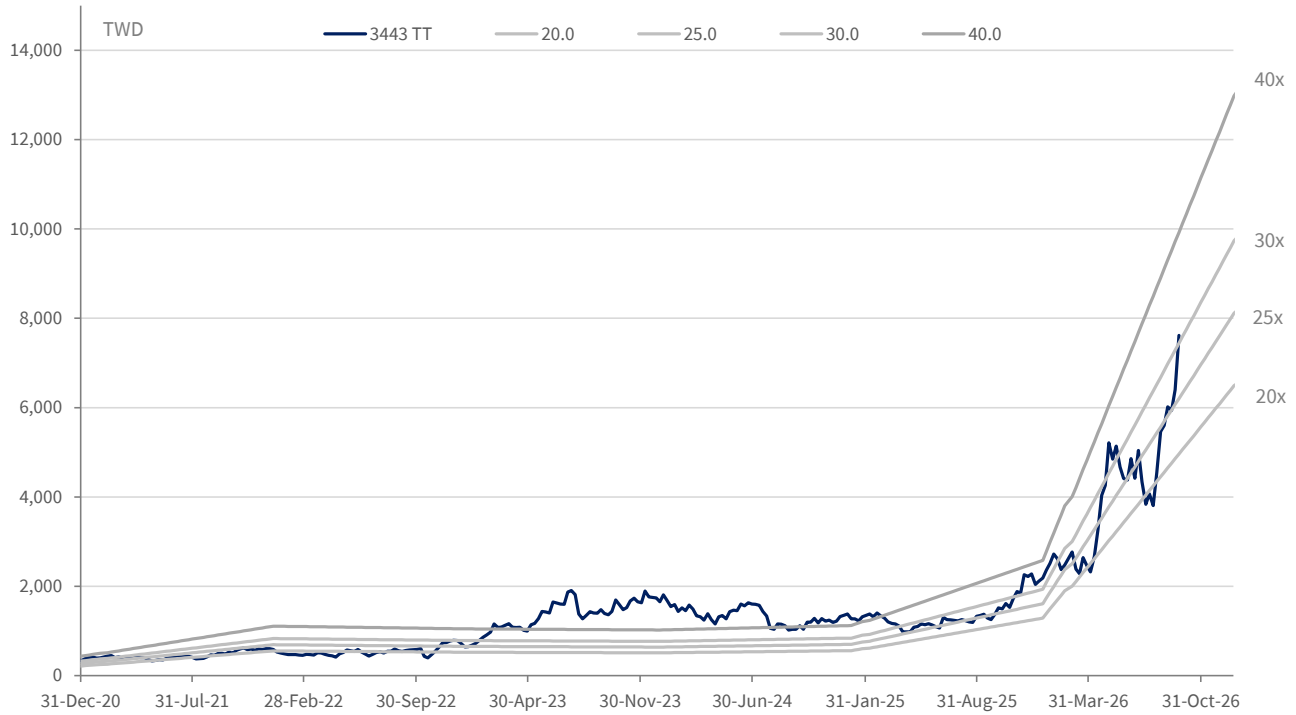
We value GUC on a PE basis, setting our TP of NT$12,000 against average FY27E/28E earnings as we move into 2H26. We use that base because we now have visibility on GUC's project engagements with every hyperscaler except Apple, and the bulk of that revenue lands in 2028E. On our EPS estimates, the TP implies 35x FY27E or 25x FY28E PER, which we still regard as undemanding.

GUC has been trading in the forward twelve-month PER range of 30x -50x. Sell-side estimates for GUC diverge widely, reflecting limited visibility on its project engagements and on the rate at which NRE converts to turnkey. We therefore apply a conservative 30x PE, even though our project-level work puts FY27-28E EPS 150-200% above consensus. Note that most of the analysts on the Street value GUC based on 40x PER. We prefer the upside and downside to sit in the earnings forecast rather than in a stretched target multiple.

Consensus and buyside whispers on GUC's FY27E EPS have drifted up from NT$150 to NT$200 over the past several months. Even so, we do not think the scale of the N3 and N5 wafer allocation, or the margin upside from a richer TK1 mix and better TK3 profitability, is yet in those numbers. Our EPS sits more than NT$100 above even the bullish whispers. The main catalyst is disclosure: GUC should reveal more engagements and turnkey projects over the next six to twelve months, having stated publicly that it holds many more (to our knowledge, more than 10 TK1 projects due to contribute in FY27E) and will disclose them as they hit milestones.

We do not use FY28 as the valuation base because projects more than a year out carry high uncertainty, even where we have some visibility, and the gap between assumption and outcome can be wide. Unlike the rest of our coverage — testing equipment and semi-consumables — GUC is therefore better valued on a next-twelve-month, or next fiscal year, basis.

FIGURE 16: GUC'S PER BAND



Source: The company, Aletheia/TAG

EARNINGS MODEL

FYE Dec, Sales mix (TWD mn)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Cloud	9,253	10,956	13,819	27,389	25,295	37,942	63,237	126,474	24,334	61,416	252,948	218,730
Consumer	1,014	1,173	1,325	1,211	1,181	1,022	1,185	950	5,534	4,723	4,338	3,943
AIoT	249	953	578	560	585	669	598	603	1,901	2,339	2,454	2,454
Automotive	427	389	951	2,037	7,608	11,411	22,823	34,234	1,170	3,804	76,075	254,868
Others	505	426	358	355	411	487	403	414	1,201	1,644	1,715	1,706
Total Consolidated Revenue	11,448	13,897	17,030	31,551	35,079	51,532	88,245	162,674	34,141	73,926	337,530	481,701
Cloud	*80.8%*	*78.8%*	*81.1%*	*86.8%*	*72.1%*	*73.6%*	*71.7%*	*77.7%*	*71.3%*	*83.1%*	*74.9%*	*45.4%*
Consumer	*8.9%*	*8.4%*	*7.8%*	*3.8%*	*3.4%*	*2.0%*	*1.3%*	*0.6%*	*16.2%*	*6.4%*	*1.3%*	*0.8%*
AIoT	*2.2%*	*6.9%*	*3.4%*	*1.8%*	*1.7%*	*1.3%*	*0.7%*	*0.4%*	*5.6%*	*3.2%*	*0.7%*	*0.5%*
Automotive	*3.7%*	*2.8%*	*5.6%*	*6.5%*	*21.7%*	*22.1%*	*25.9%*	*21.0%*	*3.4%*	*5.1%*	*22.5%*	*52.9%*
Others	*4.4%*	*3.1%*	*2.1%*	*1.1%*	*1.2%*	*0.9%*	*0.5%*	*0.3%*	*3.5%*	*2.2%*	*0.5%*	*0.4%*
Total Consolidated Revenue	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*	*100.0%*

P&L model (TWD, mn)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	11,448	13,897	17,030	31,551	35,079	51,532	88,245	162,674	34,141	73,926	337,530	481,701
COGS	-8,335	-10,903	-13,452	-25,047	-27,848	-41,064	-70,054	-129,304	-25,687	-57,736	-268,270	-388,048
Gross profit	3,113	2,994	3,579	6,504	7,231	10,468	18,191	33,371	8,454	16,190	69,261	93,654
OPEX	-1,289	-1,318	-1,521	-2,596	-2,536	-3,364	-4,790	-8,180	-4,104	-6,723	-18,870	-21,790
Operating profit	1,824	1,676	2,058	3,908	4,695	7,104	13,400	25,191	4,350	9,466	50,391	71,864
Non-op	101	80	62	88	104	133	184	288	93	330	710	879
Profit befroe tax	1,925	1,756	2,120	3,996	4,800	7,237	13,585	25,479	4,443	9,796	51,101	72,743
Tax	-278	-201	-254	-479	-576	-868	-1,630	-3,057	-673	-1,213	-6,132	-8,729
Net profit	1,646	1,555	1,865	3,516	4,224	6,369	11,955	22,421	3,770	8,583	44,969	64,014
FD EPS	***12.25***	***11.59***	***13.90***	***26.20***	***31.20***	***47.04***	***88.30***	***165.60***	***28.01***	***63.94***	***332.14***	***472.21***

Margin	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
GM	27.2%	21.5%	21.0%	20.6%	20.6%	20.3%	20.6%	20.5%	24.8%	21.9%	20.5%	19.4%
OPEX	11.3%	9.5%	8.9%	8.2%	7.2%	6.5%	5.4%	5.0%	12.0%	9.1%	5.6%	4.5%
OPM	15.9%	12.1%	12.1%	12.4%	13.4%	13.8%	15.2%	15.5%	12.7%	12.8%	14.9%	14.9%
PTM	16.8%	12.6%	12.4%	12.7%	13.7%	14.0%	15.4%	15.7%	13.0%	13.3%	15.1%	15.1%
NP	14.4%	11.2%	11.0%	11.1%	12.0%	12.4%	13.5%	13.8%	11.0%	11.6%	13.3%	13.3%

QoQ chg (%)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	-8%	21%	23%	85%	11%	47%	71%	84%				
Gross profit	32%	-4%	20%	82%	11%	45%	74%	83%				
Operating profit	45%	-8%	23%	90%	20%	51%	89%	88%				
Net profit	42%	-6%	20%	88%	20%	51%	88%	88%				
EPS	42%	-5%	20%	88%	19%	51%	88%	88%				

YoY chg (%)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	FY25A	FY26E	FY27E	FY28E
Revenue	63%	128%	98%	154%	206%	271%	418%	416%	36%	117%	357%	43%
Gross profit	58%	47%	72%	175%	132%	250%	408%	413%	4%	92%	328%	35%
OPEX	47%	21%	47%	135%	97%	155%	215%	215%	-5%	64%	181%	15%
Operating profit	67%	78%	95%	211%	157%	324%	551%	545%	14%	118%	432%	43%
Net profit	71%	99%	115%	203%	157%	310%	541%	538%	9%	128%	424%	42%
EPS	72%	99%	116%	204%	155%	306%	535%	532%	10%	128%	419%	42%

Source: Company, Aletheia/TAG

Financial statement

INCOME STATEMENT

FYE Dec. (TWD, bn)	FY23A	FY24A	FY25A	FY26F	FY27F	FY28F
Revenue	26.2	25.0	34.1	73.9	337.5	481.7
COGS	(18.3)	(16.9)	(25.7)	(57.7)	(268.3)	(388.0)
Gross profit	8.0	8.1	8.5	16.2	69.3	93.7
Operating expense	(4.0)	(4.3)	(4.1)	(6.7)	(18.9)	(21.8)
EBITDA	4.6	4.4	5.0	10.6	55.4	79.1
Operating profit	4.0	3.8	4.3	9.5	50.4	71.9
Net non-op income	0.2	0.3	0.1	0.3	0.7	0.9
Profit before tax	4.2	4.1	4.4	9.8	51.1	72.7
Tax	(0.6)	(0.6)	(0.7)	(1.2)	(6.1)	(8.7)
Minority Interest	-	-	-	-	-	-
Net Profit	3.5	3.5	3.8	8.6	45.0	64.0
FD EPS, TWD	26.2	25.6	28.0	63.9	332.1	472.2
Change						
Revenue	9.2%	-4.6%	36.3%	116.5%	356.6%	42.7%
Gross profit	-4.3%	1.7%	4.3%	91.5%	327.8%	35.2%
Operating profit	-3.2%	-4.2%	14.4%	117.6%	432.3%	42.6%
Net Profit	-5.5%	-1.6%	9.2%	127.7%	423.9%	42.4%
EPS	-5.5%	-2.3%	9.6%	128.2%	419.4%	42.2%

BALANCE SHEET

FYE Dec. (TWD, bn)	FY23A	FY24A	FY25A	FY26F	FY27F	FY28F
Cash/ST investments	9.7	13.4	8.7	11.9	13.8	20.7
Inventory	4.9	2.8	10.1	29.4	117.2	133.8
Receivable	2.0	2.0	3.4	8.1	44.2	38.0
Current assets	19.4	24.5	25.7	54.5	180.5	198.1
Fixed Assets	0.8	1.1	1.6	3.4	2.0	1.3
Goodwill & intangibles	0.6	0.4	0.6	0.6	0.5	0.5
Other LT assets	0.3	0.3	0.5	0.5	0.5	0.5
LT assets	1.6	1.8	2.7	4.4	3.0	2.3
Total assets	21.1	26.3	28.4	58.9	183.6	200.4
Payable	1.2	1.1	2.2	8.0	41.4	34.9
Other liability	9.8	13.6	12.8	21.5	23.3	25.3
Current liabilities	11.0	14.7	15.0	29.5	64.7	60.2
LT liabilities	0.4	0.3	0.5	10.5	61.1	49.8
Total liabilities	11.4	15.0	15.5	40.1	125.8	110.0
Common shares	1.3	1.3	1.3	1.3	1.3	1.3
Retained earnings	6.9	8.1	9.4	14.9	53.9	86.4
Equity	9.7	11.3	12.9	18.8	57.8	90.3
Liability&equity	21.1	26.3	28.4	58.9	183.6	200.4

CASH FLOW

FYE Dec. (TWD, bn)	FY23A	FY24A	FY25A	FY26F	FY27F	FY28F
Net profit	3.5	3.5	3.8	8.6	45.0	64.0
Dep. & amortisation	0.6	0.6	0.7	1.1	5.0	7.2
Chg in WC	1.7	1.9	-6.7	-16.6	-90.6	-16.9
Others	-1.4	0.3	1.1	5.3	1.6	1.8
OPN cash flow	4.5	6.2	-1.1	-1.6	-39.0	56.2
Capex	-0.5	-0.8	-1.2	-2.8	-3.7	-6.5
Chg in investment	-0.3	-0.8	0.7	2.0	0.0	0.0
Others	0.0	0.2	0.1	0.0	0.0	0.0
Investing CF	-0.7	-1.5	-0.4	-0.8	-3.6	-6.4
Chg in debt	-0.1	-0.1	-0.1	10.0	50.5	-11.3
Equity raised	0.0	0.0	0.0	0.0	0.0	0.0
Dividends (paid)	-1.9	-1.9	-2.1	-2.7	-6.0	-31.5
Others	-0.0	-0.0	-0.0	-0.0	0.0	0.0
Financing CF	-2.0	-2.0	-2.2	7.3	44.5	-42.7
FX & other adj.	-0.0	0.0	-0.0	0.0	0.0	0.0
Chg in cash	1.8	2.8	-3.8	4.9	1.9	7.0
Beginning cash	5.8	7.6	10.4	6.6	11.5	13.4
End cash	7.6	10.4	6.6	11.5	13.4	20.4
FCF	4.0	5.4	-2.3	-4.4	-42.6	49.7

RATIOS & PER SHARE DATA *

FYE Dec. (TWD)	FY23A	FY24A	FY25A	FY26F	FY27F	FY28F
Gross margin	30.4%	32.4%	24.8%	21.9%	20.5%	19.4%
EBITDA Margin	17.5%	17.7%	14.7%	14.3%	16.4%	16.4%
Op. Profit Margin	15.1%	15.2%	12.7%	12.8%	14.9%	14.9%
Net Profit Margin	13.4%	13.8%	11.0%	11.6%	13.3%	13.3%
ROE (%)	39.5%	32.9%	31.1%	54.1%	117.3%	86.4%
ROA (%)	16.6%	14.6%	13.8%	19.6%	37.1%	33.3%
Gross gearing	2.8%	2.5%	2.3%	54.6%	105.2%	54.8%
Net gearing	-76.3%	-90.7%	-49.4%	-6.7%	81.9%	32.2%
Asset turn (x)	1.2	1.1	1.2	1.7	2.8	2.5
Leverage (x)	2.4	2.3	2.3	2.8	3.2	2.6
Payable days	26.8	24.3	23.4	32.4	33.7	36.0
Receivable days	34.4	28.9	28.9	28.5	28.4	31.2
Inventory days	114.0	82.6	92.2	125.3	100.0	118.4
EPS	26.2	25.6	28.0	63.9	332.1	472.2
BVPS	72.2	83.7	96.0	140.4	427.0	666.4
Net cash per share	55.1	75.9	47.4	9.3	(349.9)	(214.9)
SPS	195.8	185.6	253.7	550.7	2,493.0	3,553.3
FCFPS	30.1	40.1	(17.2)	(33.1)	(314.9)	366.6
DPS	14.0	13.9	15.9	20.0	44.4	232.2

Source: Company, Aletheia/TAG

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